

**IB Mathematics SL
Internal Assessment**

This is a Sample Title

- This is a Sample Subtitle -

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1 Introduction

In this report, we explore several key aspects of mathematical writing and research communication. The goal is to practice structuring a document using $\text{\LaTeX}$, while also understanding the criteria typically used to assess mathematical investigations.

Throughout this document, we will present explanations, examples, and short exercises that you can modify while learning. We also include tables, equations, and references to other sections.¹

2 Scoring Criteria

This section summarizes the criteria commonly used to evaluate mathematical work. Each subsection describes a different scoring component.

2.1 Communication

(See this table 1 if you want)

Clear communication is essential in mathematical writing. A well-structured report should:

- present ideas in a logical sequence,
- define all mathematical symbols,
- include diagrams or tables when helpful,
- use consistent notation throughout.

As an example, consider the following table summarizing notation used in this document:

Symbol	Meaning
a_n	n -th term of a sequence
$f(x)$	Function depending on x
$\sum$	Summation notation

Table 1: Common Mathematical Symbols

2.2 Mathematical Presentation

Your work should be mathematically accurate and logically reasoned. This includes showing all steps and using appropriate formatting for equations.

¹An example footnote

Below is an example formatted using the `equation` environment:

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}. \quad (1)$$

We can also typeset systems of equations:

$$\begin{cases} x + y = 7, \\ 2x - y = 3. \end{cases}$$

2.3 Personal Engagement

Personal engagement means showing your own thinking, discoveries, or motivation. For example:

“I became curious about how the Fibonacci sequence grows, so I decided to model it using exponential approximations.”

You can also include simple experiments you perform, like generating numbers with a small script or manually exploring patterns.

2.4 Reflection

Reflection means analyzing what you have learned, what challenges you faced, and how the mathematics connects to real life.

Example reflective paragraph:

Mathematical modeling helped me understand how quickly exponential functions grow. I initially expected linear-like behavior, but after plotting several values, I realized that exponential growth is far more rapid. This changed how I interpret graphs in real-world contexts such as population models.

2.5 Use of Mathematics

This criterion assesses the correctness and depth of your mathematical work.

You can practice by adding any of the following:

- derivations,
- algebraic manipulation,
- graphs created with TikZ,

- proofs or justifications.

Here is a small example involving sequences:

$$a_n = 2n + 1 \quad \Rightarrow \quad S_n = \sum_{k=1}^n (2k + 1) = n^2 + 2n.$$

2.6 Communication

This section is repeated in your original structure, but you may keep it if needed for additional practice. Here, we include a short paragraph exercise:

Effective communication ensures the reader can follow your mathematical arguments without confusion. Using clear paragraph transitions, labeled figures, and meaningful examples helps achieve a professional writing style.

3 Conclusion

In this practice document, we created a structured $\text{\LaTeX}$ file containing sections, subsections, tables, equations, and lists. By modifying the content, adding new examples, and experimenting with formatting, you will gradually become more confident using $\text{\LaTeX}$ for academic or professional writing. [1]

Future work could involve adding bibliographies, figures, TikZ diagrams, or exploring packages such as `amsmath`, `graphicx`, and `hyperref`.

References

- [1] Alcosser, Howard. “Diamond Bar High School.” *Internal assessment: Mathematical Exploration*. Web. 26/11/2025.
- [2] Alcosser, Howard. “Diamond Bar High School.” *Internal assessment: Mathematical Exploration*. Web. 26/11/2025.
- [3] Alcosser, Howard. “Diamond Bar High School.” *Internal assessment: Mathematical Exploration*. Web. 26/11/2025.